Track 6: Open Innovation – Agriculture & Rural Development

Problem Title 1: Low-Cost Sugarcane Sett Cutting and Planting Aid

A. Problem Description

Sugarcane cultivation requires farmers to prepare and plant stem cuttings, commonly called setts. Manual cutting and planting can be time-consuming and physically demanding, particularly for small-scale farmers. Improper cutting lengths and inconsistent planting arrangements can also lead to wastage of planting material and additional labour.

Design an affordable, manually operated tool that helps farmers cut sugarcane setts to suitable lengths and prepare them for planting. The tool should prioritize operator safety, ease of maintenance, portability, and compatibility with commonly used sugarcane varieties. The design must not depend on robotic automation.

B. Challenges

1. Designing a tool that produces reasonably consistent sett lengths.
2. Reducing manual effort and preparation time.
3. Preventing injuries during cutting and handling.
4. Keeping the design affordable for small-scale farmers.
5. Making the tool durable and easy to maintain using locally available materials.

Problem Title 2: Low-Cost Paddy Field Levelling and Water-Depth Measurement Tool

A. Problem Description

Uneven paddy fields can result in inconsistent water distribution, uneven crop establishment, and difficulties during field operations. Farmers may lack affordable tools to assess field-level variations and

measure water depth at different locations before and during cultivation.

Design a simple, low-cost, manually operated tool that helps farmers assess field-level differences and measure standing-water depth at selected points in a paddy field. The tool should help farmers identify uneven areas and make better-informed decisions about field preparation and water management without requiring electronic sensors, automated control, or robotic equipment.

B. Challenges

1. Developing a practical method for identifying uneven field areas.
2. Measuring water depth consistently at different locations.
3. Making the tool easy to use in wet and muddy field conditions.
4. Keeping the device lightweight, durable, and affordable.
5. Designing the tool for different field sizes and soil conditions.

Problem Title 3: Integrated Agricultural Government Schemes and Crop Insurance Assistance Platform

A. Problem Description

Farmers may struggle to identify suitable government agricultural schemes and understand crop insurance eligibility, required documents, application procedures, premium details, claim requirements, and deadlines. Information is often distributed across different official sources, making it difficult for farmers to understand the assistance available to them or complete the necessary procedures.

Develop a user-friendly digital platform that brings together verified information about agricultural government schemes and crop insurance. The platform should help farmers identify potentially relevant schemes, understand eligibility requirements, prepare document checklists, learn how to apply, and understand the steps involved in submitting insurance claims. It should provide regional-language support and direct users to official application channels.

B. Challenges

1. Maintaining accurate and updated information about schemes and insurance policies.
2. Explaining eligibility requirements in simple language.
3. Helping farmers understand the documentation required for applications and claims.
4. Supporting regional languages and users with limited digital literacy.
5. Providing clear information about deadlines, exclusions, and claim procedures.
6. Protecting farmers' personal information and avoiding misleading assurances of eligibility or claim approval.

Problem Title 4: Banana Cultivation Cost and Harvest Planning Management System

A. Problem Description

Banana farmers must manage cultivation expenses, plantation maintenance, harvesting schedules, labour requirements, and sales planning. Without organized records, farmers may find it difficult to estimate production costs, compare expenditure across growing cycles, and plan harvesting and sales effectively.

Develop a digital application specifically for banana farmers to maintain plantation-wise records of cultivation expenses, planting dates, maintenance activities, expected harvest periods, labour requirements, and sales. The system should provide simple summaries of recorded costs and help farmers organize their cultivation and harvest plans.

B. Challenges

1. Maintaining separate records for different banana plots and planting cycles.
2. Recording expenses and farm activities with minimal effort.

3. Organizing expected harvest dates and labour requirements.
4. Helping farmers compare actual cultivation costs with sales revenue.
5. Supporting local languages and offline data entry.
6. Keeping the application simple and affordable for small-scale banana farmers.

Problem Title 5: Paddy Procurement Transparency and Digital Transaction Record System

A. Problem Description

Paddy farmers may face difficulties maintaining accurate records of the quantities sold, moisture-related quality assessments, weighing results, price deductions, and payments received from procurement centres or traders. Incomplete records can make it difficult to verify transactions, understand deductions, and resolve disagreements.

Develop a digital recordkeeping system that helps paddy farmers document their sales and procurement transactions. The application should allow farmers to record the quantity delivered, reported quality measurements, applicable prices, deductions, receipts, and payment status. It should generate easy-to-understand transaction summaries that farmers can retain for their records.

B. Challenges

1. Recording procurement transactions accurately and consistently.
2. Making price calculations and deductions easy to understand.
3. Maintaining digital copies of receipts and transaction details.
4. Supporting regional languages and farmers with limited digital literacy.
5. Protecting records and preventing accidental data loss.
6. Designing the application to work with limited internet connectivity.